

Laurent COLBOIS

Rue de Lausanne 31 Age : 23 (02.05.1996)

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Single

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Education

École Polytechnique Fédérale de Lausanne, Lausanne

Master in Computational Science and Engineering

since 2017

Bachelor in Physics

2013-2016

Lycée Denis-de-Rougemont, Neuchâtel

Federal Maturity in Physics and Applied Mathematics

2010-2013

Work experience

Logitech, Innovation Park, EPFL Campus, Lausanne

100 % (02.2019-08.2019)

Internship. Developed a deep learning model for audio environment classification, able to run in real time as an embedded system on a microcontroller. (Python, Keras, Tensorflow, C)

Supervision : Milos Cernak

École Polytechnique Fédérale de Lausanne, Lausanne

2014-2017

Student assistant in C/C++ Programming

Technical skills

Data Science

Machine learning, Deep learning

Signal Processing

Audio processing, Image processing

Numerical methods

Optimization, Computational linear algebra, Finite difference method, Finite element method

High Performance Computing

Parallel computing (OpenMP, MPI), Algorithms

Programming languages, Libraries

Python, Pytorch, Tensorflow, Keras, Matlab, C/C++, Java (Notions)

Academic projects

Course project in Audio signal processing	10 % (10.2018-12.2018)
Implemented an algorithm of speech enhancement (noise reduction), based on Wiener filtering and using LPC analysis to obtain a relevant estimate of the speech signal. (Python)	
Course project in Deep learning	10 % (04.2018-06.2018)
Implemented and evaluated various CNN architectures for binary classification of brain signals, inspired by some state-of-the-art neural networks in the field of brain-computer interface. (Python, Pytorch)	
Semester project in Computational Plasma Physics	25 % (02.2018-07.2018)
Implemented and evaluated a method aiming at computing more accurate field derivatives, in the context of a FEM simulation of a turbulent plasma. (Fortran)	
Supervision : Laurent Villard	
Semester project in Theoretical Finite Element Method	25 % (09.2018-01.2019)
Derivated a posteriori error estimates for a mixed Fourier/FEM numerical resolution of a two-dimensional PDE problem, and used them in the implementation of an adaptive-mesh algorithm. (Matlab)	
Supervision : Marco Picasso	

Languages

English : C1

French : Mother tongue

German : B2

Interests

Trumpet playing (since 2003). Diploma : *Certificat supérieur d'études non-professionnelles* (2017, Conservatoire de musique neuchâtelois)

Piano playing (since 2008). Diploma : *Certificat d'études non-professionnelles* (2017, Conservatoire de musique neuchâtelois)